Keecas Quarto Example

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How to use keecas in a Quarto document

This example demonstrates all the key features of `keecas` for symbolic and units-aware calculations in a Quarto document.

> **Note**
>
> If the rendering engine is `KaTeX` (i.e. Jupyter notebook), the label command `\label{}` will result in a `ParseError`. The latex code will work when rendering by `quarto render`, but since you may want to see the result before rendering the document, an option to disable the `\label{}` command is provided.
> When rendering with `quarto` you can have two options:
>
> - if `qmd` files, then everything is fine
> - if `ipynb` files, then pass the `--execute` flag to `quarto render` to rerender the notebook

Basic Symbolic Math with Units

Define symbols and calculate basic engineering quantities:

$$F = 10.00 \text{ kilonewton} \qquad \text{(applied force)} \tag{1}$$

$$A = 50.00 \text{ cm}^2 \qquad \text{(cross-sectional area)} \tag{2}$$

$$\sigma = \frac{F}{A} = 2.00\,\text{MPa} \qquad \text{(normal stress)} \tag{3}$$

Pipe Commands Demonstration

Using pipe operators for functional composition:

$$x = 3.00 \text{ m} \tag{4}$$

$$y = 4.00 \text{ m} \tag{5}$$

$$d = x^2 + y^2 = 25.0\,\text{m}^2 \tag{6}$$

Different LaTeX Environments

Cases Environment

$$\begin{cases} E_{steel} = 200.00 \text{ gigapascal} & \text{steel elastic modulus} \\ E_{concrete} = 30.00 \text{ gigapascal} & \text{concrete elastic modulus} \end{cases} \tag{7}$$

Equation Environment with Labels

Calculate beam deflection with cross-references:

$$\delta = \frac{5\,q\,L^4}{384\,E\,I} = 15.949\,\text{mm} \tag{8}$$

ℹ Note

labels emitted by `show_eqn` are latex labels, therefore they need to be referenced with `\eqref{}` or `\ref{}` latex commands.

The deflection calculated in 8 shows acceptable values for serviceability.

?@eq-QUARTO_EXAMPLE-delta

Verification Function

Using the `check` function for design checks:

$$\sigma_{Sd} = 20.00 \text{ MPa} \tag{9}$$

$$\tau_{Sd} = 15.00 \text{ MPa} \tag{10}$$

$$\sigma_{Rd} = 250.00 \text{ MPa} \tag{11}$$

$$\tau_{Rd} = 75.00 \text{ MPa} \tag{12}$$

$$\frac{\sigma_{Sd}}{\sigma_{Rd}} = 0.080 \qquad [\leq 1.0 \quad \textbf{VERIFICATO}] \tag{13}$$

$$\frac{\tau_{Sd}}{\tau_{Rd}} = 0.200 \qquad [\leq 1.0 \quad \textbf{VERIFICATO}] \tag{14}$$

The type of comparison in the `check` function can be specified, as well the value to be specified against.

$$a = 3 \qquad [> 1 \quad \textbf{NON VERIFICATO}] \tag{15}$$

$$b = 4 \qquad [> 2 \quad \textbf{VERIFICATO}] \tag{16}$$

$$c = 5 \qquad [\geq 3 \quad \textbf{NON VERIFICATO}] \tag{17}$$

$$d = 6 \qquad [\geq 4 \quad \textbf{VERIFICATO}] \tag{18}$$

$$f = 7 \qquad [\neq 5 \quad \textbf{NON VERIFICATO}] \tag{19}$$

Subs get ordered

Substitution will get ordered and sorted automatically, so you can use a symbol before mapping to a value.

$$
\begin{aligned}
a &= 3 && (20)\\
b &= 4 && (21)\\
d &= \frac{\sqrt{a^{2} + b^{2}}}{c} & &= 0.417 \quad (22)\\
c &= a\,b & &= 12.000 \quad (23)
\end{aligned}
$$

Summary

This example demonstrated:

- Basic symbolic math with units
- Pipe command usage (see Section)
- Different LaTeX environments (align, cases, equation)
- Label and cross-reference functionality
- Dataframe operations
- Verification functions
- Complex symbolic manipulations

All calculations in Section show that keecas provides a powerful interface for engineering calculations in Quarto documents.